

The signal–heterodyne (sig–het) likelihood for galactic binaries

WDM–domain polyphase heterodyne with a real–projection bin–fold

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Abstract

This note documents the mathematics of the *signal–heterodyne* (sig–het) galactic–binary (GB) likelihood implemented in `gbsignalhetcomputations.py` / `gb_signal_het_wdm_v2.py` and the backing C++ kernel `gb_signal_het_get_ll_in_kernel`. The method evaluates the Wilson–Daubechies–Meyer (WDM) wavelet–domain log–likelihood of a near monochromatic GB without ever forming the dense WDM template. It rests on three ideas: (i) a *heterodyne* factorization of the candidate template against a fixed reference, (ii) a *polyphase* per–layer minimal–length inverse FFT that produces the candidate WDM coefficients only on a sparse time grid, and (iii) a *linearized bin–fold* that pre–contracts the data and reference into per–bin coefficients so each likelihood call is a handful of cheap sums. We also derive the *real–WDM projection* that removes the spurious $\sim 10^{-3}$ accuracy floor present in the original complex/Hermitian bin–fold.

Contents

1	The WDM–domain likelihood	2
2	The complex (quadrature) WDM transform	2
3	Heterodyne factorization	3
4	Sparse polyphase evaluation of the candidate	5
5	Linearized bin–fold of the inner products	5
6	Real–WDM projection and the inner products	6
6.1	The complex–vs–real subtlety	6
6.2	Real projection of $\langle d h \rangle$	7
6.3	Real projection of $\langle h h \rangle$	7
7	Assembled likelihood and algorithm	8
8	Validation summary	8

1 The WDM-domain likelihood

LISA TDI data $d_c(t)$ in channels $c \in \{X, Y, Z\}$ (or $\{A, E, T\}$) are sampled at cadence Δt over $N = N_f N_t$ samples. The Gaussian log-likelihood for a template $h_c(\boldsymbol{\theta})$ is

$$\ln \mathcal{L}(\boldsymbol{\theta}) = \langle d | h \rangle - \frac{1}{2} \langle h | h \rangle - \frac{1}{2} \langle d | d \rangle, \quad (1)$$

with a noise-weighted inner product. We work in the *WDM wavelet domain*: a real time series $x_c(t)$ is mapped to real coefficients w_{cmn}^x on a time-frequency lattice indexed by a frequency layer m and a time pixel n . The layer spacing is $\Delta f = 1/(2N_f \Delta t)$ and the pixel spacing is $\Delta \tau = N_f \Delta t$. In this basis the inner product is diagonal in (m, n) and reduces to a sensitivity-weighted sum,

$$\langle a | b \rangle = \sum_{m,n} \sum_{c,c'} w_{cmn}^a (\mathbf{C}^{-1})_{cc'mn} w_{c'mn}^b, \quad (2)$$

where $(\mathbf{C}^{-1})_{cc'mn}$ is the inverse noise covariance (the inverse of the per-pixel cross-channel sensitivity matrix; for diagonal $\{A, E, T\}$ it collapses to a scalar $(\mathbf{C}^{-1})_{cmn}$). All overall factors (4dc, quadrature weights) are folded into $(\mathbf{C}^{-1})$ so that (2) is exact against the `lisatools.template_likelihood`.

Why this is hard for a GB. A galactic binary is nearly monochromatic: its power sits in $\lesssim 5$ frequency layers around the carrier f_0 , but spread across *all* N_t time pixels. A brute-force likelihood forms the dense w_{cmn}^h for every candidate $\boldsymbol{\theta}$ via a length- N_t inverse FFT per layer — the dominant cost in an MCMC. The sig-het method removes that cost.

2 The complex (quadrature) WDM transform

It is convenient to use the *complex* (quadrature) WDM transform, whose coefficients $\mathcal{C}_{cmn}^x \in \mathbb{C}$ carry both the cosine- and sine-phase of the wavelet response. Writing $\hat{x}_c(k) = \sum_l x_c(l) e^{-2\pi i k l / N}$ for the length- N DFT of the (windowed) channel, the coefficient at layer m , pixel n is

$$\mathcal{C}_{cmn}^x = \kappa s_{mn} \bar{\mathcal{C}}_{mn} \sqrt{\Delta t} \underbrace{\frac{1}{N_t} \sum_{j=-N_t/2}^{N_t/2-1} \tilde{\Phi}(j) \hat{x}_c\left(m \frac{N_t}{2} + j\right) e^{2\pi i j n / N_t}}_{\equiv G_{cmn}^x}, \quad (3)$$

with the normalisation and phase factors

$$\kappa = \frac{2\sqrt{\pi}}{N_f}, \quad s_{mn} = (-1)^{(m+1)n}, \quad \bar{\mathcal{C}}_{mn} = \begin{cases} 1, & m+n \text{ even}, \\ -i, & m+n \text{ odd}, \end{cases} \quad (4)$$

and $\tilde{\Phi}$ the Meyer analysis window

$$\tilde{\Phi}(\omega) = \frac{1}{\sqrt{\Delta\Omega}} \times \begin{cases} 1, & |\omega| < A, \\ \cos\left[\frac{\pi}{2} I_{p,p}\left(\frac{|\omega|-A}{B}\right)\right], & A \leq |\omega| < A+B, \\ 0, & \text{otherwise,} \end{cases} \quad \begin{aligned} \Delta\Omega &= \pi/N_f, \\ A &= \Delta\Omega/4, \quad B = \Delta\Omega - 2A, \end{aligned} \quad (5)$$

evaluated at $\omega = 2\pi j/N$, where $I_{p,p}$ is the regularised incomplete beta function with $p = 4$. The window has support $|j| < N_t/2$ FD bins, so the natural gather length is exactly N_t . The layer centre maps to DFT bin $m N_t/2$ because the layer frequency $m\Delta f$ corresponds to bin $m\Delta f (N\Delta t) = m N_t/2$.

The bridge to the real likelihood. The real WDM coefficient used in (2) is exactly the real part of the complex one,

$$w_{cmn}^x = \text{Re } \mathcal{C}_{cmn}^x \quad (6)$$

term for term. Equation (6) is the linchpin of Section 6: the likelihood is a quadratic form in $\text{Re } \mathcal{C}$, *not* in $|\mathcal{C}|$.

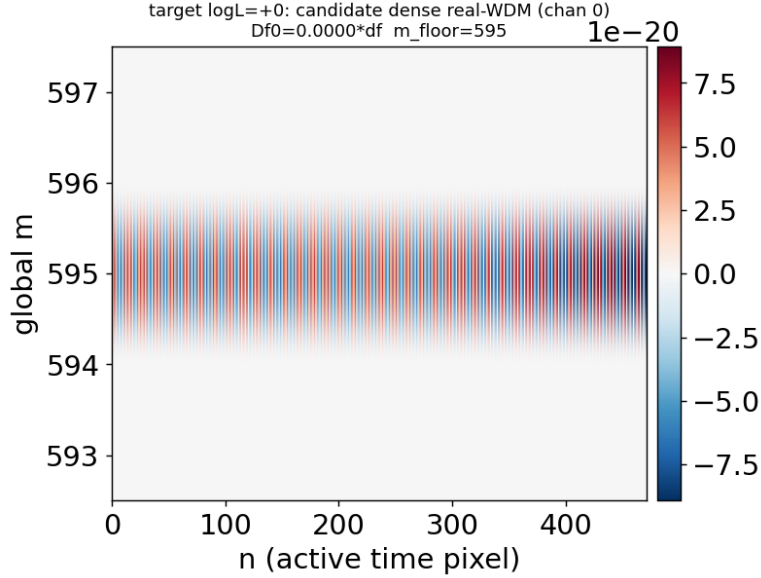


Figure 1: The (real) WDM coefficients $w_{cmn} = \text{Re } \mathcal{C}_{cmn}$ of a galactic binary on the active frequency band $m_0 \pm M$ ($M = 2$), for the highest-frequency mojito source ($f_0 = 20.38$ mHz, channel X). Power sits in the carrier layer m_0 and its two neighbours and runs across every time pixel n — this is the object the dense likelihood forms per candidate, and the one the sig-het method avoids building.

3 Heterodyne factorization

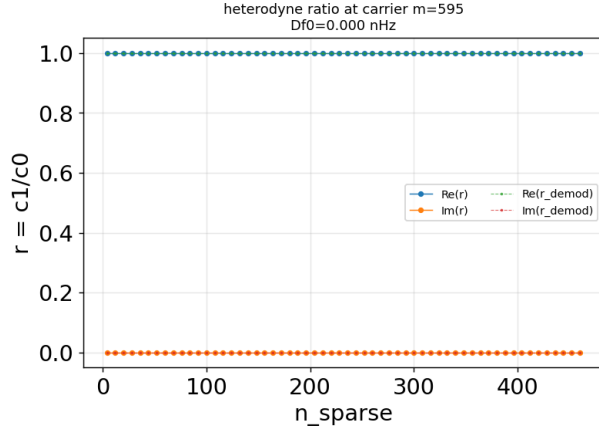
Fix a *reference* parameter vector θ_0 (in practice the catalogue source, or the current MCMC mode) and precompute its dense complex WDM coefficients once,

$$c_{cmn}^0 \equiv \mathcal{C}_{cmn}^{h(\theta_0)}. \quad (7)$$

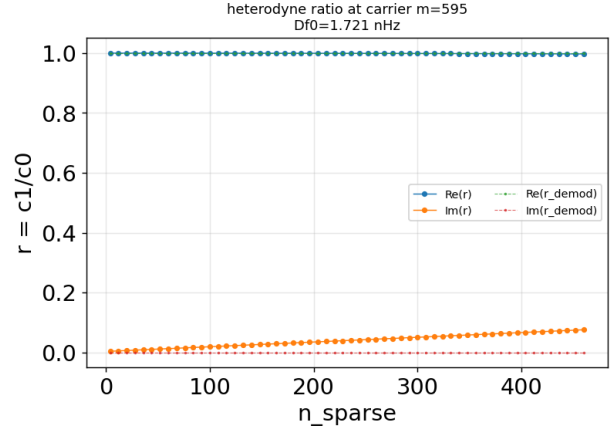
For a candidate θ with complex WDM $c_{cmn}^1 \equiv \mathcal{C}_{cmn}^{h(\theta)}$, define the per-pixel *heterodyne ratio*

$$r_{cmn} \equiv \frac{c_{cmn}^1}{c_{cmn}^0} \quad (\text{where } |c_{cmn}^0| \text{ is above a small floor}). \quad (8)$$

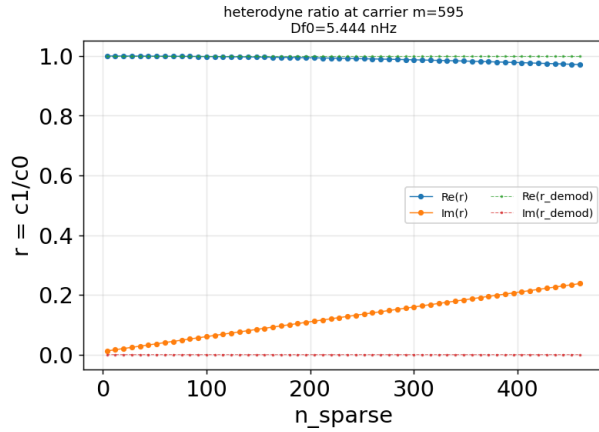
For a candidate close to the reference, r_{cmn} is a slowly varying function of the time pixel n within each layer: the rapid pixel-to-pixel oscillation of the GB carrier is shared by c^1 and c^0 and cancels in the ratio, leaving a smooth low-order phase/amplitude drift (a near-linear wind in n set by the carrier mismatch Δf_0). This is the property the bin-fold exploits. The candidate template is recovered as $c_{cmn}^1 = r_{cmn} c_{cmn}^0$, i.e. $h_{cmn} = r_{cmn} c_{cmn}^0$.



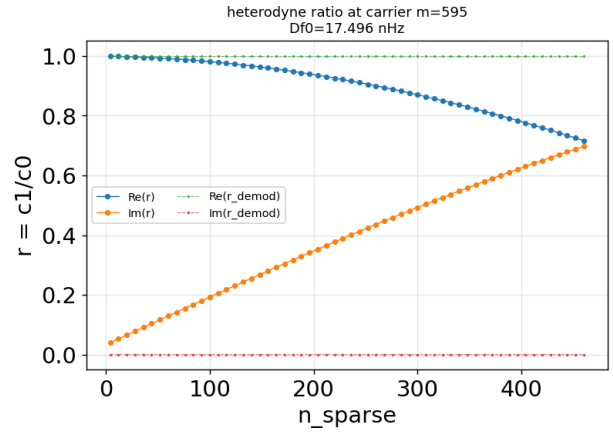
(a) $\ln \mathcal{L} = 0$ (candidate = reference): $r \equiv 1$.



(b) $\ln \mathcal{L} \approx -5$: gentle wind.



(c) $\ln \mathcal{L} \approx -50$.



(d) $\ln \mathcal{L} \approx -500$: full half-turn.

Figure 2: The heterodyne ratio $r_{cm}(n) = c^1/c^0$ at the carrier layer $m = m_0$, as the candidate f_0 is walked away from the reference (the four sig-het rows of the explainer, mojito rank 0). At the peak $r \equiv 1$; off the reference it becomes a smooth, near-linear *wind* in n set by the carrier mismatch Δf_0 (blue/orange = Re / Im r). The de-rotated $r_{\text{demod}} = r e^{-2\pi i \Delta f_0 t_n}$ (green/red) removes that ramp and stays flat, confirming the slow variation the linearized bin-fold (13) exploits.

4 Sparse polyphase evaluation of the candidate

We only ever need c^1 on the active frequency layers $m \in \mathcal{M} = \{m_0 - M, \dots, m_0 + M\}$ (with $m_0 = \lfloor f_0/\Delta f \rfloor$ and half-width M , default $M = 2$) and on a *sparse* time grid

$$n_b = \nu + b\sigma, \quad b = 0, \dots, N_b - 1, \quad \sigma = N_t/N_\ell, \quad (9)$$

i.e. every σ -th pixel, with σ the *stride* and N_ℓ the per-layer FFT length. The inner factor G_{cmn}^x in (3) is a length- N_t inverse DFT; evaluating it only on the strided grid lets us replace it by a length- N_ℓ inverse DFT.

Polyphase identity. Write the windowed, pre-phased FD samples as

$$W_j = \tilde{\Phi}(j) \hat{x}_c\left(m\frac{N_t}{2} + j\right), \quad j = 0, \dots, N_t - 1 \quad (10)$$

(absorbing the centring shift into the index). We want $G(n) = \frac{1}{N_t} \sum_{j=0}^{N_t-1} W_j e^{2\pi i j n/N_t}$ at $n = \nu + b\sigma$. Split $e^{2\pi i j n/N_t} = e^{2\pi i j \nu/N_t} e^{2\pi i j b\sigma/N_t}$ (using $\sigma/N_t = 1/N_\ell$) and absorb the first exponential into a *pre-phase*, $\tilde{W}_j = W_j e^{2\pi i j \nu/N_t}$. Decomposing $j = qN_\ell + \rho$ with $q = 0, \dots, \sigma - 1$ and $\rho = 0, \dots, N_\ell - 1$, and using $e^{2\pi i q N_\ell b\sigma/N_t} = 1$,

$$G(\nu + b\sigma) = \frac{1}{N_t} \sum_{\rho=0}^{N_\ell-1} \underbrace{\left(\sum_{q=0}^{\sigma-1} \tilde{W}_{qN_\ell+\rho} \right)}_{Y_\rho} e^{2\pi i \rho b\sigma/N_t} = \frac{1}{\sigma} \text{IDFT}_{N_\ell}[Y]_b. \quad (11)$$

So: *fold* the length- N_t pre-phased window into length N_ℓ by summing the σ polyphase branches, run *one* length- N_ℓ inverse FFT, and scale by $1/\sigma$. This is mathematically exact for any σ dividing N_t — the only approximation in the whole method is the bin-fold linearization of Section 5, never the polyphase step. The per-layer cost drops from $\mathcal{O}(N_t \log N_t)$ to $\mathcal{O}(N_\ell \log N_\ell)$.

Restoring the WDM dressing of (3), the sparse candidate coefficient is

$$c_{cmn_b}^1 = \sqrt{\Delta t} \kappa s_{mn_b} \bar{c}_{mn_b} \frac{1}{\sigma} \text{IDFT}_{N_\ell}[Y^{(c,m)}]_b. \quad (12)$$

The Hermitian wrap $\hat{x}_c(-k) = \hat{x}_c^*(k)$ (real time series) supplies the negative-frequency window bins; the reference c^0 is computed once by the ordinary dense transform.

Choosing N_ℓ . Polyphase is exact at any valid N_ℓ , but the stride $\sigma = N_t/N_\ell$ sets the spacing of the sparse grid on which r is sampled. N_ℓ is auto-calibrated at construction by increasing $N_\ell \in \{16, 32, 64, \dots\}$ until the sparse output matches the dense lisatools reference to a per-pixel relative difference $< 10^{-12}$, and so that r is well represented by a line over σ pixels (typically $\sigma \lesssim 16$).

5 Linearized bin-fold of the inner products

The inner products (2) are sums over *all* pixels n , but we have r only at the sparse centres n_b . Within bin b (the pixels n closest to n_b) we linearise the smooth ratio,

$$r_{cmn} \approx r_{cm}(n_b) + \dot{r}_{cm}(n_b)(n - n_b), \quad \dot{r}_{cm}(n_b) = \frac{r_{cm}(n_{b+1}) - r_{cm}(n_{b-1})}{2\Delta n}, \quad (13)$$

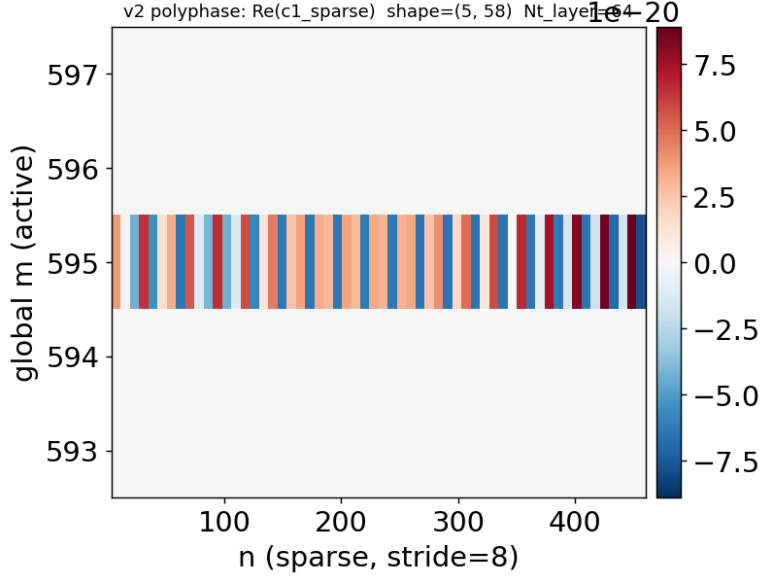


Figure 3: The sparse complex-WDM coefficients $\text{Re } c_{cmn_b}^1$ produced by the polyphase step (11) for the same source, on the active layers (5 rows) and the strided time grid ($\sigma = 8$, so $N_b = 58$ columns). A single length- N_ℓ inverse FFT per layer replaces the length- N_t dense transform of Fig. 1; the result is exact (per-pixel relative difference $< 10^{-12}$) at the sparse pixels.

with $\overline{\Delta n}$ the mean bin width (one-sided differences at the ends). The template enters the inner products only through $h_{cmn} = r_{cmn} c_{cmn}^0$, so substituting (13) lets us pre-contract everything that does *not* depend on θ — the data d , the reference c^0 , and (C^{-1}) — into per-bin coefficients. This is the key move: the expensive all-pixel sums are done *once*, at construction; each likelihood call only sums over the N_b sparse bins.

We give the contraction directly in its real-projection form (Section 6 explains why this, and not the naive complex form, is correct).

6 Real-WDM projection and the inner products

6.1 The complex-vs-real subtlety

By (6) the likelihood is a quadratic form in $\text{Re } \mathcal{C}$. A natural but *wrong* shortcut is to evaluate the Hermitian complex form

$$\frac{1}{2} \text{Re} \sum_{cc'mn} \mathcal{C}_{cmn}^{a*} (C^{-1})_{cc'mn} \mathcal{C}_{cmn}^b = \frac{1}{2} \sum_{cc'mn} (C^{-1})_{cc'mn} (\text{Re } \mathcal{C}^a \text{Re } \mathcal{C}^b + \text{Im } \mathcal{C}^a \text{Im } \mathcal{C}^b), \quad (14)$$

which carries a spurious $\text{Im} \cdot \text{Im}$ term relative to the true real form $\sum \text{Re } \mathcal{C}^a (C^{-1}) \text{Re } \mathcal{C}^b$. The two agree only when $\sum \text{Im}^2 \approx \sum \text{Re}^2$ by phase averaging, which holds for the broadband $\langle d | d \rangle$ but *not* pixel-by-pixel in the bin-fold. The residual

$$\frac{1}{2} \sum (C^{-1}) (\text{Im } \mathcal{C}^a \text{Im } \mathcal{C}^b - \text{Re } \mathcal{C}^a \text{Re } \mathcal{C}^b) \quad (15)$$

is exactly the $\sim 6.5 \times 10^{-3}$ effective-mismatch floor observed in the original sig-het. The fix is to project each coefficient to its real part *before* contracting.

6.2 Real projection of $\langle d|h \rangle$

Write $u_{cmn} = \text{Re } c_{cmn}^0$, $v_{cmn} = \text{Im } c_{cmn}^0$, so that with $h = r c^0$,

$$\text{Re } h_{cmn} = u_{cmn} \text{Re } r_{cmn} - v_{cmn} \text{Im } r_{cmn}. \quad (16)$$

The (real) data–template term is, using (2),

$$\langle d|h \rangle = \sum_{mn} \sum_{cc'} \text{Re } d_{cmn} (\mathbf{C}^{-1})_{cc'mn} \text{Re } h_{c'mn} = \sum_{mn} \sum_{c'} D_{c'mn}^{\text{Re}} (u_{c'mn} \text{Re } r_{c'mn} - v_{c'mn} \text{Im } r_{c'mn}), \quad (17)$$

with the data pre-contraction $D_{c'mn}^{\text{Re}} = \sum_c \text{Re } d_{cmn} (\mathbf{C}^{-1})_{cc'mn}$ (a scalar $\text{Re } d_{cmn} (\mathbf{C}^{-1})_{cmn}$ in the diagonal case). Inserting the linearization (13) and bin-folding gives, per channel c' , layer m , bin b , the two real moment pairs

$$A_0^{\text{Re}} = \sum_{n \in b} D^{\text{Re}} u, \quad A_0^{\text{Im}} = \sum_{n \in b} D^{\text{Re}} v, \quad A_1^{\text{Re}} = \sum_{n \in b} D^{\text{Re}} u (n - n_b), \quad A_1^{\text{Im}} = \sum_{n \in b} D^{\text{Re}} v (n - n_b). \quad (18)$$

The crucial repacking is to store these as *single complex* coefficients

$$\boxed{A_0 = 2(A_0^{\text{Re}} + i A_0^{\text{Im}}), \quad A_1 = 2(A_1^{\text{Re}} + i A_1^{\text{Im}}),} \quad (19)$$

because then the *existing* complex kernel contraction reproduces the real value at no extra storage:

$$\frac{1}{2} \text{Re} \sum_{c'mb} (A_0 r + A_1 \dot{r}) = \sum_{c'mb} (A_0^{\text{Re}} \text{Re } r - A_0^{\text{Im}} \text{Im } r + A_1^{\text{Re}} \text{Re } \dot{r} - A_1^{\text{Im}} \text{Im } \dot{r}) = \langle d|h \rangle. \quad (20)$$

The middle identity uses $\frac{1}{2} \text{Re}[2(A_0^{\text{Re}} + i A_0^{\text{Im}})(\text{Re } r + i \text{Im } r)] = A_0^{\text{Re}} \text{Re } r - A_0^{\text{Im}} \text{Im } r$.

6.3 Real projection of $\langle h|h \rangle$

The template norm is genuinely quadratic in r , so a single complex coefficient cannot absorb it; two blocks are needed. Using (16),

$$\langle h|h \rangle = \sum_{mn} \sum_{cc'} (\mathbf{C}^{-1})_{cc'mn} \text{Re } h_{cmn} \text{Re } h_{c'mn}, \quad (21)$$

and the algebraic identity (with $h = r c^0$, valid for symmetric $(\mathbf{C}^{-1})$)

$$2 \text{Re } h_c h_{c'} = \underbrace{c_c^{0*} (\mathbf{C}^{-1}) c_{c'}^0}_{\text{conj}} r_c^* r_{c'} + \underbrace{c_c^0 (\mathbf{C}^{-1}) c_{c'}^0}_{\text{non-conj}} r_c r_{c'} \quad (22)$$

(after multiplying by $(\mathbf{C}^{-1})$ and summing) motivates the *two* bin-folded blocks

$$B_0 = \sum_{n \in b} c_c^{0*} (\mathbf{C}^{-1})_{cc'} c_{c'}^0, \quad B_1 = \sum_{n \in b} c_c^{0*} (\mathbf{C}^{-1})_{cc'} c_{c'}^0 (n - n_b), \quad (\text{conj}) \quad (23)$$

$$B_0^{\text{nc}} = \sum_{n \in b} c_c^0 (\mathbf{C}^{-1})_{cc'} c_{c'}^0, \quad B_1^{\text{nc}} = \sum_{n \in b} c_c^0 (\mathbf{C}^{-1})_{cc'} c_{c'}^0 (n - n_b), \quad (\text{non-conj}) \quad (24)$$

where for diagonal $\{A, E, T\}$ the channel pair (c, c') collapses to one index. The kernel then forms

$$\boxed{\langle h|h \rangle = \frac{1}{2} \text{Re} \sum_{c'cb} \left[B_0 r_c^* r_{c'} + B_0^{\text{nc}} r_c r_{c'} + B_1 (r_c^* \dot{r}_{c'} + \dot{r}_c^* r_{c'}) + B_1^{\text{nc}} (r_c \dot{r}_{c'} + \dot{r}_c r_{c'}) \right].} \quad (25)$$

To see that (25) is the real norm, note (suppressing m, b and writing $r_c = a_c + ib_c$, $c_c^0 = u_c + iv_c$, so $h_c = c_c^0 r_c$):

$$B_0 r_c^* r_{c'} + B_0^{\text{nc}} r_c r_{c'} = (\mathbf{C}^{-1})_{cc'} (h_c^* h_{c'} + h_c h_{c'}) = (\mathbf{C}^{-1})_{cc'} h_{c'} (2 \text{Re } h_c), \quad (26)$$

and taking $\frac{1}{2} \text{Re}$ leaves $\sum (\mathbf{C}^{-1})_{cc'} \text{Re } h_c \text{Re } h_{c'} = \langle h | h \rangle$ (the B_1 blocks carry the same identity for the linearization slope). The non-conjugated blocks are the only new storage versus the original Hermitian form. Dropping the $\dot{r}^2$ (second-order, $(n - n_b)^2$) term keeps $\langle h | h \rangle$ to the standard linear-in- r budget off the reference — exact at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ and tracking the dense WDM likelihood to $\lesssim 10^{-2}$ in $\ln \mathcal{L}$ even hundreds of units away from the peak. A flag `project_real` selects this path; with it disabled the kernel reproduces the old complex form (14) bit for bit.

7 Assembled likelihood and algorithm

Collecting (20) and (25), the log-likelihood is

$$\ln \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2} \text{Re} \sum_{c'mb} (A_0 r + A_1 \dot{r}) - \frac{1}{2} \cdot \frac{1}{2} \text{Re} \sum_{cc'mb} [\dots]_{(25)} - \frac{1}{2} \langle d | d \rangle, \quad (27)$$

with $\langle d | d \rangle$ a fixed scalar from the (real) lisatools data inner product. The two heterodyne terms are returned by the kernel as $\langle d | h \rangle$ and $\langle h | h \rangle$; the frontend `GBSignalHetComputations.get_ll` adds $-\frac{1}{2} \langle d | d \rangle$ and returns $\ln \mathcal{L}$ directly.

Per likelihood call (the only $\boldsymbol{\theta}$ -dependent work).

1. Synthesize the candidate FD signal $\hat{h}_c(\boldsymbol{\theta})$ (fast FD-direct GB generator, or TD synth + `rfft`).
2. For each active layer $m \in \mathcal{M}$: gather the N_t windowed FD bins, apply the pre-phase, polyphase-fold to length N_ℓ , one inverse FFT, dress with $\kappa s_{mn} \bar{C}_{mn} \sqrt{\Delta t}$ — giving $c_{cmn_b}^1$ [Eq. (11)].
3. Form $r = c^1 / c^0$ and $\dot{r}$ at the sparse centres [Eqs. (8), (13)].
4. Contract against the precomputed $A_0, A_1, B_0, B_1, B_0^{\text{nc}}, B_1^{\text{nc}}$ to get $\langle d | h \rangle, \langle h | h \rangle$ [Eqs. (20), (25)], then (27).

Once, at construction (independent of $\boldsymbol{\theta}$). Build the data WDM d and reference $c^0 = \mathcal{C}^h(\boldsymbol{\theta}_0)$ on the active band; compute $\langle d | d \rangle$; bin-fold $D^{\text{Re}}, u, v, (\mathbf{C}^{-1})$ into $A_0, A_1, B_0, B_1, B_0^{\text{nc}}, B_1^{\text{nc}}$.

8 Validation summary

- **Polyphase** reproduces the dense lisatools complex WDM to $< 10^{-12}$ per pixel at any calibrated N_ℓ (it is algebraically exact).
- **Real projection** lowers the effective mismatch floor from 6.5×10^{-3} (Hermitian) to $\sim 2 \times 10^{-11}$ (kernel `project_real=1` vs. `=0`).
- **End-to-end vs. mojito data** (top three high- f GBs, $N_t = 512$; Table 1): the sig-het $\ln \mathcal{L}$ tracks the dense lisatools $\ln \mathcal{L}$ to $\mathcal{O}(10^{-9})$ at the injection and $\lesssim 5 \times 10^{-2}$ out at $\ln \mathcal{L} = -500$, matching the chunked-heterodyne path. Narrowband mismatches at the peak: $\text{mm5} \sim 3 \times 10^{-11}$.

Table 1: Absolute deviation $|\ln \mathcal{L}_{\text{sig-het}} - \ln \mathcal{L}_{\text{base}}|$ of the signal-het likelihood from the dense lisatools likelihood, for the three highest-frequency mojito galactic binaries, with the candidate f_0 walked to four target $\ln \mathcal{L}$ values (Fig. 2 shows the ratios for rank 0). Last column: narrowband mm5 at the peak. All likelihoods are taken from the installed `get_ll/template_likelihood` methods.

Source (rank, f_0 , SNR)	$ \ln \mathcal{L}_{\text{sig-het}} - \ln \mathcal{L}_{\text{base}} $ at target $\ln \mathcal{L} =$				mm5 _{peak}
	0	-5	-50	-500	
0 (20.38 mHz, 57.6)	2.5×10^{-8}	2.9×10^{-4}	2.9×10^{-3}	2.3×10^{-2}	3.3×10^{-11}
1 (19.67 mHz, 23.8)	1.6×10^{-9}	4.8×10^{-4}	3.9×10^{-3}	8.2×10^{-3}	3.9×10^{-11}
2 (16.72 mHz, 61.1)	5.2×10^{-9}	5.9×10^{-4}	5.8×10^{-3}	5.3×10^{-2}	2.4×10^{-11}

Edge handling. Both the sig-het and chunked paths use a WDM time-edge cut $\geq$ the Tukey taper width $\lceil \alpha N_t / 2 \rceil$ layers, so their active time region matches the dense reference exactly; a mismatched cut biases the absolute $\ln \mathcal{L}$ by a fixed fraction (different time coverage), not the per-pixel floor that the real projection removed.